



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	OPTOELECTRONICS AND SEMICONDUCTOR DEVICES				
Type of Course	Major Elective (SPECIALIZATION I: MATERIALS SCIENCE/SPECIALIZATION II: PHOTONICS)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY5EJ302(1)- Materials Science				
Course Summary	The Optoelectronics and Semiconductor Devices course focuses on equipping students with an understanding of the principles, operation, design, and applications of optoelectronic devices and semiconductor devices.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding the operation and characteristics of various optoelectronic devices, such as light-emitting diodes (LEDs), laser diodes, etc	U	F	Instructor-created exams / Quiz
CO2	Understanding of semiconductor physics, including band theory, carrier transport, and semiconductor device operation principles.	U, R	F	Instructor-created exams / Quiz

CO3	Understand the knowledge about the radiative transition processes and other optoelectronic phenomenon.	U, Up	F	Instructor-created exams / Quiz
CO4	Understand the applications of optoelectronic and semiconductor devices in various fields such as telecommunications, imaging, sensing, displays, and energy conversion.	U, An	F	Instructor-created exams / Quiz
CO5	To be familiar with equipment and devices that work on the principle of semiconducting phenomena and theories of optoelectronics	U, Ap	F	Instructor-created exams / Quiz
CO6	Students will gain hands-on experience through laboratory experiments involving the characterization and testing of optoelectronic and semiconductor devices	An, Ap	F	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	AN INTRODUCTION TO OPTOELECTRONICS		12	18
	1	Emission and absorption processes, Photon statistics	2	
	2	The behaviour of electrons, Optical properties of some common materials	3	
	3	Electrons in a periodic lattice, Metals, insulators and semiconductors	3	
	4	Refraction, Absorption and emission, Fluorescence, Scattering	2	
	5	The absorption and emission of light by semiconductors	2	
	References: Section A1:1.6- 1.8, 2.2,2.6-2.10 of Book 1			

II	SEMICONDUCTOR SCIENCE AND LIGHT-EMITTING DIODES		16	22
	6	Semiconductor Science and Light-Emitting Diodes, energy Band diagrams.	3	
	7	The density of States, fermi-dirac function and metals, Extrinsic Semiconductors: n-type and p-type Semiconductors	3	
	8	compensation doping, nondegenerate and degenerate Semiconductors, Energy Band Diagrams in the Applied field (Basic ideas only: derivations not required)	4	
	9	Direct Band and Indirect band semiconductors: E-K Diagrams, PN-Junction principles (Basic ideas only: derivations not required)	3	
	10	Open circuits, PN Junction forward and reverse circuits (Basic ideas only: derivations not required)	3	
References: Section 3.1 to 3.6 of Book 2				
III	OPTOELECTRONIC SEMICONDUCTOR DEVICES I		10	15
	11	Visible light-emitting diodes : Physics of LEDs, Optical properties of LEDs.	4	
	12	Radiative and non-radiative recombination, Electrical properties,	3	
	13	Current-voltage characteristics, Efficiencies, High efficiency LEDs and novel technologies	3	
Reference : B1.1 Visible light-emitting diode of Book 1				
IV	OPTOELECTRONIC SEMICONDUCTOR DEVICES II		10	15
	19	Stimulated emission devices: Stimulated emission and Population inversion.	2	
	20	Photon amplification and laser Principles, Stimulated emission and einstein coefficients	2	
	21	Principle of the laser diode, hetero structure laser diodes	3	
	22	Photovoltaic devices: Solar cell: Basic Principles, operating current and Voltage and fill factor.	3	
References: Section 4.1.A-B, 4.2.A,4.9,4.10.5.14. A-B of Book 2				
V	OPEN ENDED MODULE		12	
		Interferometers, Thin Film Optics: Multiple Reflections in Thin Films, LED Electronics, Equivalent Circuit of a Solar Cell, Solar Cell Structures and Efficiencies, Solar cell driving a load, Open circuit voltage and short circuit current		

	Sections 1.13-1.15, 3.18, 5.14.D of Book 2		
Books and References:			
1. Handbook of Optoelectronics Volume II, John P Dakin & Robert G W Brown, 2006 by Taylor & Francis Group (Book 1)			
2. Optoelectronics and Photonics: Principles and Practices S.O. Kasap, Pearson (Book 2)			
3. Physics of Optoelectronics, Michael A. Parker, 2005 by Taylor & Francis Group, LLC (Book 3)			
4. Optics and Photonics: An Introduction, SECOND EDITION ,F. Graham Smith et al, John Wiley and Sons (Book 4)			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	2	2	3	2	3	2	3	2	3	3	2
CO 2	3	2	2	2	3	2	3	2	3	2	3	3	1
CO 3	3	2	2	2	3	2	3	2	3	2	3	3	1
CO 4	3	2	2	2	3	2	3	2	3	2	3	3	2
CO 5	3	2	2	2	3	2	3	2	3	2	3	3	3
CO 6	3	3	3	3	3	3	3	2	3	2	3	3	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	